NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
Department of Computer Science & Engineering
End-Semester Examination, Spring 2022-23
B Tech 6^{th} Semester

Subject Name: Computer Networks **Maximum Marks: 50**
Subject Code: CS3002 **Exam Duration: 3 Hours**
Number of pages: 2

Note: There are ***Six*** questions. Attempt ***All*** the questions. All parts of a same question should be attempted in a sequence.

Q.1 (a) Describe the different types of resource records used in DNS.
(b) Identify the in-band, out-band, stateless property observed in the following protocols: HTTP, FTP, DNS, and SMTP.
(c) Describe the internet routing protocols RIP and BGP.
(d) Consider the bit string 1011011001, compute the hamming code using even parity bit. Also compute the CRC code using generator polynomial $x^4 + x^2 + 1$.
(e) Suppose 1024KByte of application data (including application layer header) has to be sent out of a host attached to Ethernet. Assume the application uses TCP protocol and that the IP as well as TCP header is 20 bytes each. Further assume that Ethernet MTU size is 1500 Bytes and TCP uses segments of size 1480. What is the size (in bytes) of the last datagram? Express answer in bytes.

[2+2+2+2+2=10 Marks]

Q.2 Consider TCP Reno protocol to examine the behavior of congestion window size. Initially, (Transmission round, Window size) is (1, MSS). Assuming threshold *ssthresh* is set after first timeout. If first time out is occurred at 6^{th} transmission round (starts from 1^{st}). Determine the following by providing a short discussion justifying your answer:
(a) Show the diagram for window size up to 26^{th} transmission round; assume that first, 3 duplicate ACKs are detected at transmission round 18^{th}. No loss event occurred till 26^{th} transmission round, thereafter.
(b) What are the values of *ssthresh* at the 17^{th} and 20^{th} transmission rounds, respectively?
(c) What is average throughput for transmission round between 19^{th} to 26^{th}. Assume that MSS and RTT are 512 Bytes and 200 msec, respectively.

[2+2+4=8 Marks]

Q.3 (a) Consider the initial RTO is 6s; Seg 1 sent at 0s, Seg 2 sent at 2s, ACKs for Seg 1 and Seg 2 are received at 5s, Seg3 is sent at 8s, Seg 4 is sent at 10s, ACKs for Seg3 and Seg4 are lost, Seg3 and Seg4 are retransmitted after timeout, Seg5 is sent at 45s, ACKs for Seg3 and Seg4 received at 50s, ACK for Seg5 received at 60s. Calculate the final RTO by using α=0.125 and β=0.25.
(b) Consider sending an object of size O = 128 Kbytes from server to client. Let S=512 bits and RTT=50msec. Suppose the transport protocol uses static windows with window size W. For a transmission rate of 1Mbps, determine the minimum possible latency. Determine the minimum window size that achieves this latency.

[4+4=8 Marks]

Q.4 Suppose an organization owns the class C network 191.116.0.0/24. This organization has Six departments: three of approx. 120 nodes (named as A, B, C), two of 48 nodes (D, E), and one of 200 nodes (F). Router R is responsible for forwarding packets to the rest of the Internet, and that the routers are connected in the pattern A-B-C-D-R-E-F.
(a) Design an efficient arrangement of subnet addresses and masks to assign to each department. Also show the starting and ending addresses for each subnet.
(b) How much larger can each subnet grow in terms of nodes if you exclude the subnet address and broadcast address? What happens if several smaller departments (Four departments of 48 nodes each) are added? Just explain from where in the address space you would allocate these subnets.

[4+4=8 Marks]

Q.5 (a) Six thousand airline reservation stations are competing for the use of a single slotted ALOHA channel. The average station makes 20 requests /hour. A slot is 120 μsec. What is the approximate total channel load? What is the expected number of transmission attempts needed?

(b) Suppose nodes X and Y are on the same 100 Mbps Ethernet segment, and the propagation delay between the two nodes is 200 bit times. Suppose X sends frame at t=0 and Y sends frame at t=100 bit times, the frames collide, and then X and Y choose different values of K in the CSMA/CD algorithm. At what time, nodes X and Y detect collisions. Suppose K_X=1 and K_Y=0. Can the retransmissions from X and Y collide? At what time does X schedule its retransmission? At what time does Y begin transmission? At what time does Y's signal reach X? Does X refrain from transmitting at its scheduled time? Assuming no other nodes are active and frame size is 1000 bits. Explain the answer with sequence timing diagram.

[2+6=8 Marks]

Q.6 Consider the network scenario and costs/delay to each link as shown in Figure 1. Assume that each node knows their routing table initially [Use Distance Vector Routing].

(a) Now link between nodes R5 and R6 is broken then update the routing table at node R5 as per the distance given in Figure 1. Show all the steps involved for computing the routing table at R5.

(b) In continuing to part (a), if node R5 receives the distance vectors from node R1, R2, and R8 (0,3,5,7,6,7,8,5), (4,0,6,7,6,7,8,5), and (4,9,5,7,6,7,8,0) respectively. Node R5 knows the costs to its neighbors. Determine the routing table at node R5.

(c) Now, remove all the nodes except R1, R2, R3, and R4. Suppose R1 is connected through an internet router G. Show the delay in routing tables at R1, R2, R3 and R4 with respect to router G respectively in the following cases: (Use Hop count metric)

- If an internet router is down initially then an internet router is up. Show the routing tables up to 3 exchanges.
- If an internet router is down again then show the routing tables up to 4 exchanges.

[2+2+4=8 Marks]

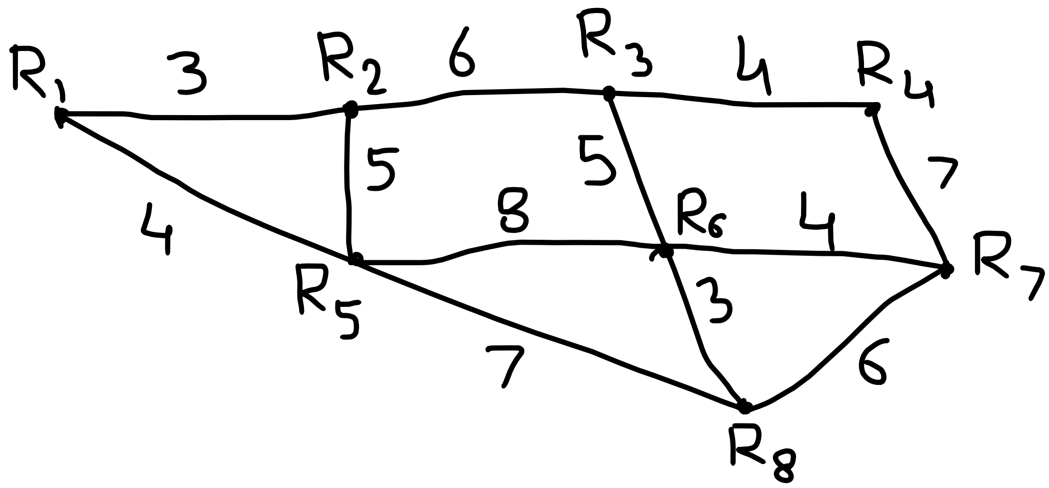

Figure 1